

Science - 1

DNA - long polymer - 2 strands of complementary information
 - Deoxy ribonucleic acid - consists of Adenine, Guanine, Thymine, Cytosine and Uracil.

Order of Magnitude

↳ factors of 10 (usually singular)

Scientific notation → representing quantities $\%d \cdot 10^n$

Length:-

SI unit → metres

Range of universe = 10^{40} → 40 orders of magnitude

Observable universe $\sim 10^{26}$ m.

Proton $\sim 10^{-15}$ m

Time:-

SI unit → ~~sec~~ second

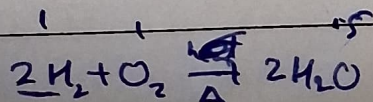
Planck Time → 10^{-44} s

Age of Universe → 10^{17} s

Range → 60 orders

Energy:-

SI unit → Joule



Chemical bond $\sim 10^{-19}$

Supernova $\sim 10^{44}$

Range $\sim 10^{63}$

Constant Rate Law:-

$$\text{Let } \frac{dy}{dx} = c$$

$$\Rightarrow dy = c dx$$

$$\Rightarrow \int dy = \int c dx$$

$$\Rightarrow \cancel{y(x)} = \cancel{\lambda} + c x$$

$$y(x) = cx + \lambda$$

Here, $c \rightarrow \text{slope}$, $\lambda \rightarrow \text{intercept}$

State Dependent Growth

$$\text{Let } \frac{dy}{dx} = cy$$

$$\int dy = \int cx dx$$

$$y = \frac{cx^2}{2} + \lambda$$

$$\frac{dy}{dx} = cy$$

$$\int \frac{1}{y} dy = \int c dx \Rightarrow cx = e \ln y + \lambda$$

$$\frac{x-\lambda}{c} = \ln y \Rightarrow y = e^{\frac{(x-\lambda)c}{e}} = \frac{e^{cx}}{e^{\lambda}} = p e^{cx}$$

What are vector fields?

classmate

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$$\text{Pressure} = \frac{\frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}}{\text{s}^2}}{\frac{\text{m}^2}{\text{s}^2}} = \frac{\text{kg} \cdot \text{m} \cdot \text{s}^{-2}}{\text{m}^2}$$

Expansion to complex no's

Let $c = a + ib$

$$y = p e^{cx} = p e^{ax + ibx} = p e^{ax} e^{ibx}$$

This models growing oscillations bounded by exponential envelopes $\pm p e^{ax}$

But if $a < 0$, the oscillations converge

Now to compute double differentials easily?

$$\text{Let } x_1 = x ; x_2 = \frac{dx}{dt}$$

$$\vec{R} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \vec{R}' = \frac{d\vec{R}}{dt} = \begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} x_2 \\ -2x_1 \end{bmatrix}$$

$$\frac{d\vec{R}}{dt} = g(\vec{R})$$

↳ vector fields

Non-Dimensionalisation

Removal of physical units from eqns by substituting characteristic scale variables

Like representing Pressure in basic variables

1) Identify vars

2) Define scale factors

3) Replace vars in terms of scale factors

Why are LC circuits damped oscillators? ^{ideal} classmate

Lotka - Volterra
Volterra

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~~Lotka - Volterra~~ Model - Prey - predator model.

1 prey species & 1 predator species

The pop of foxes & pop of rabbits.

We can build a continuous time model

Rate Laws

1) Rabbits alone

Let $x(t)$ be the pop. of the rabbits

Then $\frac{dx}{dt} \propto x(t)$

→ too hard to predict - must compute

reproduction.

$x \rightarrow x+x$

$$\Rightarrow \frac{dx}{dt} = \alpha_b x \quad \hookrightarrow \text{can be a fn of } x \text{ itself.}$$

$$\Rightarrow x \sim e^{\alpha_b t}$$

death

$x \rightarrow \emptyset$

$$\Rightarrow \frac{dx}{dt} = \alpha_d x \Rightarrow x \sim e^{\alpha_d t}$$

Total system:-

$$\frac{dx}{dt} = \underbrace{(\alpha_b - \alpha_d)}_{\hookrightarrow \text{let } \alpha} x = \lambda e^{(\alpha_b - \alpha_d)t}$$

2) Rabbits + Foxes

Let $y(t)$ be the population of the foxes.

reproduction
 $y \rightarrow y + y$

$$\frac{dy}{dt} \propto$$

roughly how many interactions
 b/w the two.

$$x y$$

①

poor approx since $x \rightarrow \phi$

death
 $y \rightarrow \phi$

$$\frac{dy}{dt} \propto$$

$$y$$

②

Total System:-

①

$$\frac{dy}{dt} =$$

$$\gamma x y$$

$$- \delta y$$

②

Hence
 This

②

$$\frac{dx}{dt} =$$

$$\alpha_b x$$

$$- \beta x y$$

$$- \alpha_d x$$

$$=$$

$$\alpha x - \beta x y$$

$\alpha, \beta, \gamma, \delta$ cannot be obtained theoretically - need fieldwork.

But how do you solve the system described by ① and ②?

One way is the Euler's difference scheme.

Approximate $\frac{df}{dt}$ as $\frac{f(t+\Delta t) - f(t)}{\Delta t}$ and then simplify.

Only hold when $\Delta t \rightarrow 0$ though.

The maxima & the minima of the prey just precedes the ones for the predator.

vector field.

$$\text{But let } \vec{R} = \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \vec{R}' = \vec{F}(\vec{R})$$

$\Rightarrow$ a focus for the vector field; magnitude $\uparrow$ away from the focus.

This is a steady / stationary state

$$\vec{F}(\vec{r}) = 0$$

$$\Rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha x - \beta xy \\ \gamma xy - \delta y \end{bmatrix}$$

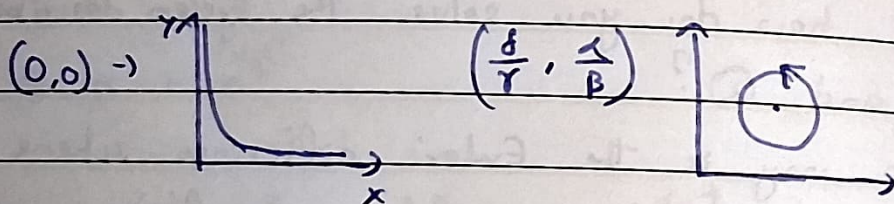
$$\alpha x = \beta xy \rightarrow y = \frac{\alpha}{\beta} x$$

$$\gamma x \frac{\alpha}{\beta} = \delta \frac{\alpha}{\beta} \quad x = \frac{\gamma}{\delta} \frac{\delta}{\gamma}$$

$$2 \text{ solns} = (0, 0); \left(\frac{\delta}{\gamma}, \frac{\alpha}{\beta} \right)$$

~~first component~~

Null lines



Detour Start:

$$\text{Take } f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \dots$$

↳ Taylor Series expansion

Higher order expansion $\rightarrow$ better approximation.

0th order = pt; 1st order = line; 2nd order = parabola

Stationary states $\rightarrow$ first derivative is zero

Detour End:-

Take $f(x, y)$

$$f(x, y) = f(x_0, y_0) + \frac{\partial f(x_0, y_0)}{\partial x} (x - x_0)$$

$$+ \frac{\partial f(x_0, y_0)}{\partial y} (y - y_0)$$

$$+ \frac{\partial (\partial f / \partial x)}{\partial x} (x - x_0)^2 + \frac{\partial (\partial f / \partial x)}{\partial y} (x - x_0)(y - y_0)$$

$$+ \frac{\partial (\partial f / \partial y)}{\partial x} (x - x_0)(y - y_0) + \frac{\partial (\partial f / \partial y)}{\partial y} (y - y_0)^2$$

$$\vec{F}(\vec{r}) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} f_1(x_0, y_0) \\ f_2(x_0, y_0) \end{bmatrix} + \begin{bmatrix} \partial f_1 / \partial x \\ \partial f_1 / \partial y \end{bmatrix} (x - x_0)$$

$$+ \begin{bmatrix} \partial f_2 / \partial x \\ \partial f_2 / \partial y \end{bmatrix} (y - y_0) + \dots$$

$$= A + \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}$$

→ Jacobian J

At stationary state, $A = 0$

$$\vec{r}' = \vec{r} - \vec{r}_0 \Rightarrow \frac{d\vec{r}'}{dt} = \vec{J}_0 \vec{r}'$$

eigenvalues

$$\vec{J} \vec{R} = \lambda \vec{R} \rightarrow \text{eigenvectors.}$$

$$J_k = \begin{bmatrix} \alpha - \beta\gamma & -\beta\gamma \\ \gamma\gamma & \gamma\alpha - \delta \end{bmatrix}$$

$$J(0,0) = \begin{bmatrix} \alpha & 0 \\ 0 & -\delta \end{bmatrix}$$

Supposition $\Rightarrow$ Ansatz (wot?) $:- \vec{R}(t) = c_1(t) \vec{v}_1 + c_2(t) \vec{v}_2$

$$\vec{R}'(t) = \frac{dc_1}{dt} \vec{v}_1 + \frac{dc_2}{dt} \vec{v}_2 \quad (1)$$

c_1, c_2 time dependent
coeffs since R is
time dependent

$$\begin{aligned} \vec{J} \vec{R}' &= J(c_1 \vec{v}_1 + c_2 \vec{v}_2) \\ &= c_1 J \vec{v}_1 + c_2 J \vec{v}_2 \\ &= c_1 \lambda_1 \vec{v}_1 + c_2 \lambda_2 \vec{v}_2 \quad (2) \end{aligned}$$

From (1) & (2)

$$c_1' = \lambda_1 c_1 \quad \& \quad c_2' = \lambda_2 c_2$$

$\Downarrow$

$$c_1 = c_1(0) \cdot e^{\lambda_1 t} \quad c_2 = c_2(0) \cdot e^{\lambda_2 t}$$

$$\lambda_1 > 0 \quad \lambda_2 < 0 \quad \text{hence}$$

$$c_1 \uparrow \quad \& \quad c_2 \downarrow$$

Perform the same calculation for the other stationary point

The Logistic Map:-

x_{n+1} = Map :- Mathematical fn that predicts the state of a system

$$\bullet x_{n+1} = f(x_n)$$

$$\bullet \text{ Given } x_0; f(x_0) = x_1; f(f(x_0)) = f(x_1) = x_2 \dots$$

We used Euler Integration

$$x_{n+1} = x_n + f(x_n) \cdot \Delta t \quad \text{where } f' = \text{differentiation}$$

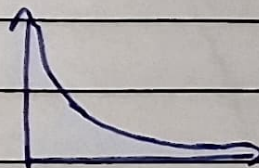
Case 1:- $f(x)$ is ~~and~~ constant

We just get $x_0, c, c, c, \dots$

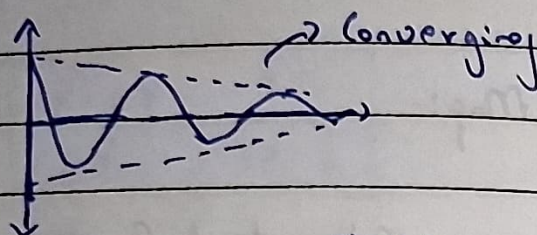
Case 2:- $f(x) = mx$

We get $x_0, mx_0, m^2x_0, m^3x_0, \dots$

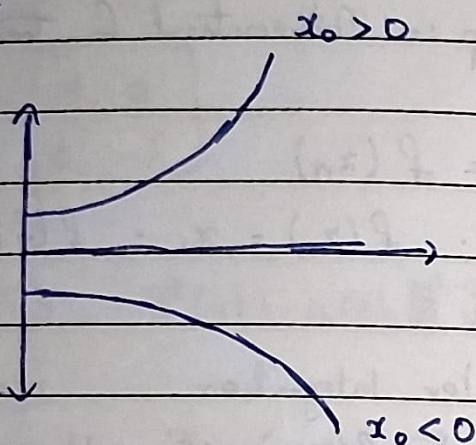
(i) $m \in (0, 1)$



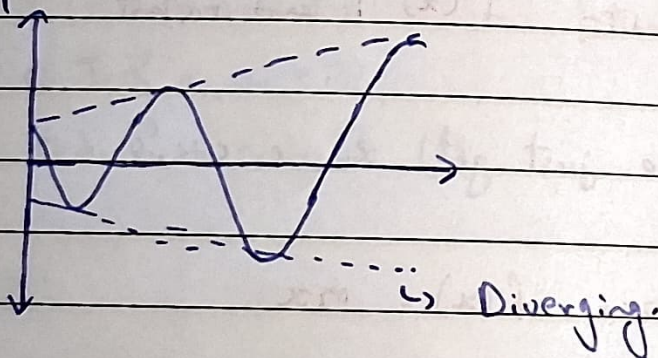
(ii) $m \in (-1, 0)$



(iii) $m > 1$

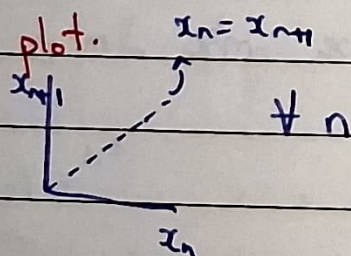


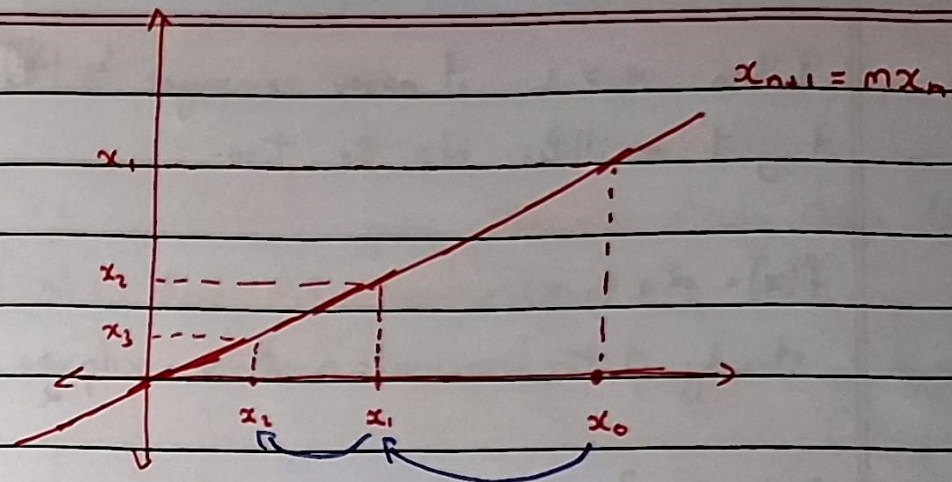
(iv) $m \leq -1$



Cobweb plot.

Plot





As we can see, since $m < 1$ & > 0 , the x_n values converge to 0.

Try plotting for other values of m too

Also what happens ~~if~~ when $mx+c$

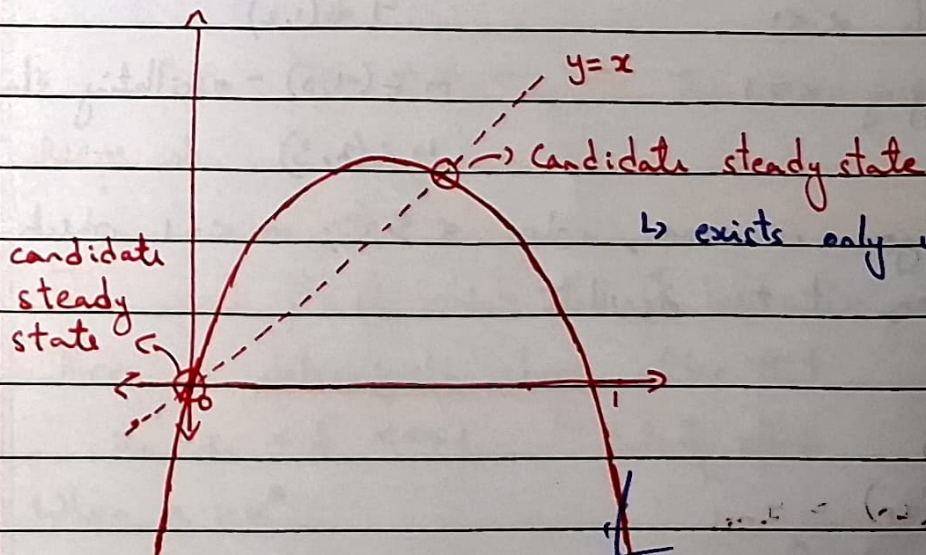
Multi variable linear maps??

Case 3: $f(x) = ax^2 + bx + c$

More simply $f(x) = ax(1-x)$

→ Logistic Map.

→ Show that these two are equivalent



When $\alpha > 3$, it never converges to the steady, instead it just oscillates b/w the two.

$$f(x) = \alpha x(1-x)$$

$$\text{steady states} \Rightarrow x = 0 \text{ \& \; } x = 1 - \frac{1}{\alpha}$$

But why?

Let's linearise.

$$x^* = 0$$

$$x^* = 1 - \frac{1}{\alpha}$$

$$f(x) = \alpha x - \alpha x^2$$

$$f(x) = f(x^*) + f'(x^*)(x-x^*) + \dots \quad (\text{Taylor expansion})$$

$$f(x) = x^* + (\alpha - 2\alpha x^*)(x - x^*)$$

$$= x^* + \alpha(1 - 2x^*)(x - x^*)$$

$$\approx C + mx$$

$$f(x) = x^* + (\alpha - 2\alpha x^*)(x - x^*) \quad \& \; m = 2 - \alpha$$

$$f(x) = 0 + (\alpha - 0)(x - 0)$$

$$= \alpha x$$

$m \in (0, 1)$ - monotonically stable

Stable $\alpha < 1$

$\hookrightarrow \alpha \in (1, 2)$

Diverging $\alpha > 1$

$m \in (-1, 0)$ - oscillating stable

$\hookrightarrow \alpha \in (2, 3)$

As you can see, when $\alpha > 3$; $m < -1$ which causes it to oscillate

$$\alpha > 3$$

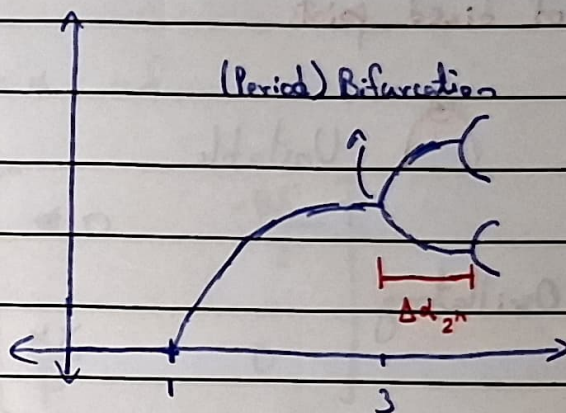
$$f(x_n) = x_{n+1}$$

$$f(x_{n+1}) = x_{n+2} = x_n$$

→ from observations (2-cycle existence)

$$x = f(f(x)) \mid x_a, x_b$$

At $\alpha > 3$, $f(f(x))$ gains two new steady states (slope < 1) while $f(x)$'s steady state becomes unstable
 ↳ 2 complex solutions turn real.



Each bifurcation moves the cycle to higher 2^i cycles

Period Doubling:- When one stationary state becomes unstable, 2 new ones appears

$\Delta \alpha_{2^n} \gg \Delta \alpha_{2^{n+1}} \Rightarrow$ The window before bifurcation keeps ↓

→ α^*

$\exists$ a real pt at which the bifurcation reaches 2^∞

There is deterministic chaos after that.

$$x_0' - x_0 = \delta \ll 1 \quad \text{↳ Butterfly effect.}$$

When $\alpha > \alpha^*$

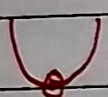
$$x_n - x_n' = \varepsilon \quad \text{where } \varepsilon \sim \delta^n \text{ when } \alpha < \alpha^* \text{ but}$$

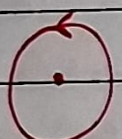
Tut-1

Fixed Points:- Points where $\frac{dy}{dx} = 0$

We use Jacobians for analysing these fixed point
 ↳ through eigenvalue decomposition

Three types of fixed points

Stable  Unstable 

 Oscillatory

$$J = \begin{bmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{bmatrix}$$

$$F_1 = \alpha x - \beta xy$$

$$F_2 = \gamma xy - \delta y$$

$$= \begin{bmatrix} \alpha - \beta y & -\beta x \\ \gamma y & \gamma x - \delta \end{bmatrix}$$

set both to
zero

at $(0,0)$ & $\left(\frac{\delta}{\gamma}, \frac{\alpha}{\beta}\right)$

(i) $\begin{bmatrix} \alpha & 0 \\ 0 & -\delta \end{bmatrix}$

$$J - \lambda I = \begin{bmatrix} \alpha - \lambda & 0 \\ 0 & -\delta - \lambda \end{bmatrix}$$

$$\lambda = \alpha, -\delta$$

(ii) $\begin{bmatrix} \alpha & 0 & -\frac{B\delta}{\gamma} \\ \frac{\alpha\gamma}{\beta} & 0 \end{bmatrix}$

$$J - \lambda I = \begin{bmatrix} -\lambda & -\frac{B\delta}{\gamma} \\ \frac{\alpha\gamma}{\beta} & -\lambda \end{bmatrix} = 0$$

$$\lambda^2 + \frac{B\delta\alpha}{\gamma} = 0$$

$$\lambda = -\frac{B\delta}{\gamma}, \quad \lambda = \pm i\sqrt{\frac{B\delta\alpha}{\gamma}}$$

Unstable fixed point $\rightarrow$ ^{both eigenvalues} ~~one eigenvalue is +ve, the other is -ve~~

Oscillating fixed point $\rightarrow$ both eigenvalues are ^{imaginary} ~~complex~~

Stable fixed point $\rightarrow$ both eigenvalues are complex

Passing a fixed pt into a map returns the same point.

Modelling pop:- $N_{T+1} = Y N_T \left(1 - \frac{N_T}{K}\right)$

Classical Mechanics.

Temp is a ~~scalar~~ scalar field - diff values for each point in the room.

Laws of Newton

1) Inertia :- Unless acted upon by an external unbalanced force, ~~the~~ body remains in a state of rest or in a state of uniform motion.

↳ Cannot be verified empirically

2) Momentum :- Rate of change of momentum of an object is the ~~new~~ net force acting on an object
interaction ~~creating~~ forces along the inter-particle axis -

3) Force :- For central forces, every action has an equal and opposite reaction (vectorially)

1) Given ~~the~~ $\vec{F} = 0$

$\vec{v}$ = constant 3d vector

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

$$\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} = \text{constants}$$

$$2) \vec{p} = m\vec{v}$$

$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a}$$

3) Given objects A, B

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

if $\vec{F}$ is a central force

Superposition Principle

Given an n-particle system,

$$\vec{F}_A = \sum_{\substack{i=1 \\ i \neq A}}^{n-1} \vec{F}_{Ai}$$

In presence of an electric and a magnetic field

$$\vec{F}_A = q(\vec{E} + \vec{v} \times \vec{B}) \quad \text{if A has a charge } q.$$

Work & Energy:

$$W = \int_{t_a}^{t_b} \vec{F}(t) \cdot d\vec{r}(t) = \int_{t_a}^{t_b} \frac{d\vec{p}(t)}{dt} \cdot \vec{v}(t) dt$$

$$= \int_{t_a}^{t_b} m \frac{d\vec{v}(t)}{dt} \cdot \vec{v}(t) dt = \int_{t_a}^{t_b} m \vec{v}(t) \cdot d\vec{v}(t)$$

$$= \frac{1}{2} m v^2 \Big|_A^B$$

$$= KE_A - KE_B \text{ where Kinetic Energy} = \frac{1}{2} m v^2$$

Conservative Forces - Gradient of a scalar field

Central forces are conservative in nature.

$$U(\vec{r}) \text{ s.t. } \vec{F} = \left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z} \right)$$

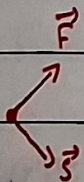
$$W = \int \vec{F} \cdot d\vec{r} = \int -\nabla U d\vec{r} = -(U_B - U_A)$$

$$K_B - K_A = U_A - U_B$$

$$K_A + U_A = K_B + U_B \rightarrow \text{Law of Conservation of Energy}$$

Law of conservation of momentum:-

When no external unbalanced force acts on an object, its velocity is constant.



$$\text{Given } \vec{F}(t) \cdot \vec{S} = 0$$

$$\Rightarrow \frac{d\vec{p}}{dt} \cdot \vec{S} = 0 \Rightarrow \frac{d\vec{p} \cdot \vec{S}}{dt} = 0$$

Linear momentum is conserved for directions $\perp$ to the force.

Moment of a force:-

Torque:- If $\exists$ an ideal rod and a force is applied at a point r away, then $\vec{\tau} = \vec{r} \times \vec{F}$. It is the moment of force at a point r ~~at a point~~

Angular Momentum:-

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$

$$\frac{d\vec{r} \times \vec{p}}{dt} = \frac{d m \vec{r} \times \vec{v}}{dt} = m \vec{r} \frac{d\vec{v}}{dt} = \vec{I} \vec{a} = \vec{\tau}$$

$$\text{Let } E(t) = K(t) + U(t)$$

$$\frac{dE(t)}{dt} = 0 \rightarrow \text{conservation of energy}$$

$$\text{Let } f(x_1, x_2, \dots, x_n, t)$$

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n$$

$$+ \frac{\partial f}{\partial t} dt$$

$$= \nabla f \cdot d\vec{r} + \frac{\partial f}{\partial t} dt$$

Similarly

$$\frac{d\vec{E}}{dt} = \vec{F}(x, p, t)$$

$$= \frac{p^2}{2m} + U(x)$$

$$\frac{dE}{dt} = \frac{\partial E}{\partial x} dx + \frac{\partial E}{\partial p} dp + \frac{\partial E}{\partial t} dt$$

$$= \frac{\partial U}{\partial x} dx + \frac{\partial K}{\partial p} dp + \frac{\partial U}{\partial t} dt$$

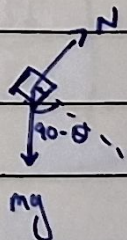
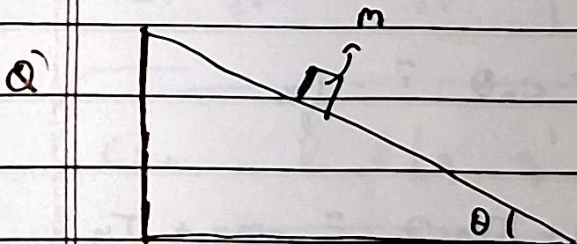
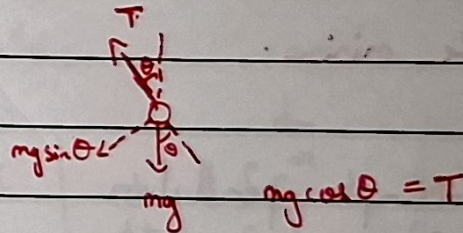
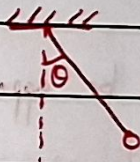
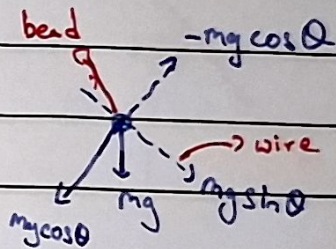
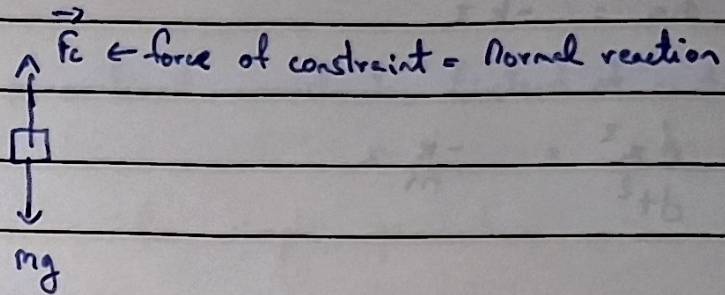
=

$$\frac{dE}{dt} = \frac{\partial U}{\partial t}$$

$$\text{Hence } \frac{dE}{dt} = 0 \text{ iff } \frac{\partial U}{\partial t} = 0$$

$$\text{If } \exists \text{ non conservative forces } \Rightarrow \frac{\partial U}{\partial t} \neq 0$$

Net force = forces of interaction + forces of constraints.



$$N = mg \cos \theta$$

$$F = m$$

Q) $F = -kx$

$$ma = -kx$$

$$\frac{dx^2}{dt^2} = \frac{-k}{m} x$$

$$a + \omega^2 x = 0$$

$$\text{where } \omega^2 = \frac{k}{m}$$

$$E = \frac{1}{2} kx^2 + \frac{1}{2} mv^2$$

Even other harmonic motions can be approximated to SHO near the minima.

In a simple pendulum

$$ma_x = \vec{F}_{\text{net}x} = -T \sin \theta \approx -\frac{T_x}{l}$$

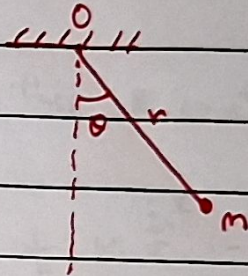
$$ma_y = \vec{F}_{\text{net}y} = -mg + T \cos \theta \approx -mg + \frac{T_y}{l}$$

Lagrangian Mechanics:

$$\text{Lagrangian } L = KE - PE$$

Use of generalised coordinates (instead of cartesian coordinates)

Polar coordinates for simple pendulum



$$\frac{m dr^2}{dt^2}$$

Euler - Lagrange Equations

$$\frac{d}{dt} \frac{\delta L}{\delta \dot{r}'} = \frac{\delta L}{\delta r} \quad \left| \quad \frac{d}{dt} \frac{\delta L}{\delta \dot{\theta}'} = \frac{\delta L}{\delta \theta}$$

Given a fn $f(q, q', x)$

$$\frac{\delta f}{\delta q} = \frac{d}{dx} \left(\frac{\delta f}{\delta q'} \right)$$

$$\frac{d}{dt} L = \frac{1}{2} m (\dot{\theta})^2 + mgl \cos \theta$$

$$L(\theta, \dot{\theta})$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = \frac{d}{dt} m l \ddot{\theta} = mgl (-\sin \theta)$$

$$l \ddot{\theta} = -g \sin \theta$$

when $\theta \rightarrow 0$

$$l \ddot{\theta} = -g \theta$$

Two body problem:-

$$\vec{r}_1 = (x_1, x_2, x_3)$$

$$\vec{r}_2 = (x_4, x_5, x_6)$$

$$L = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - U(|\vec{r}_1 - \vec{r}_2|, m_1, m_2)$$

$$\text{Centre of mass} \rightarrow \underbrace{m_1 \vec{r}_1 + m_2 \vec{r}_2}_{m_1 + m_2} = \vec{R} = \frac{m_1 \vec{r}_1 + m_2 (\vec{r}_1 - \vec{r})}{m_1 + m_2}$$

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

KE of Com

$$L = -U(|\vec{r}|) + \underbrace{\frac{1}{2} M \dot{R}^2}_{\downarrow \text{KE of Com}} + \underbrace{\frac{1}{2} \mu \dot{r}^2}_{\downarrow \mu = \frac{m_1 m_2}{m_1 + m_2}}$$

$$\text{Now, } L = f(\vec{R}, \dot{\vec{R}}, t, \vec{r}, \dot{\vec{r}})$$

simplifies to $f(\vec{r}, \dot{\vec{r}}, t, \vec{R})$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\vec{R}}} - \frac{\partial L}{\partial \vec{R}} = 0$$

Hence $\frac{\partial L}{\partial \dot{\vec{R}}} = \text{const} = m \dot{\vec{R}}$

Let's set this const to 0. (changing frame to CM)

$$L = \frac{1}{2} \mu \dot{\vec{r}}^2 - U(|\vec{r}|)$$

Choose the plane have the ^{basis} $\vec{r}$ & $\vec{v}$

$$\dot{\vec{r}}(r, \theta, \phi)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} = \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{\partial L}{\partial \dot{\phi}} = \text{const} = 0 \text{ (since } \phi = 0 \text{)}$$

$$\dot{\vec{r}} = \frac{dr}{dt} \hat{r} + \omega \hat{v}$$

$$L = \frac{1}{2} \mu \dot{\vec{r}}^2 - U(r) = \frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\theta}^2) - U(r)$$

Now $L = f(r, \dot{r}, \dot{\theta})$

E-L eqn for θ

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = \frac{\partial L}{\partial \theta} = 0 \Rightarrow \frac{\partial L}{\partial \dot{\theta}} = \text{const} = \mu r^2 \dot{\theta} = l$$

$$\frac{1}{2} \mu \cdot r \cdot \dot{\theta}^2 = \text{Areal velocity} = \text{constant.}$$

E.L. eqn for r

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = \frac{\partial L}{\partial r}$$

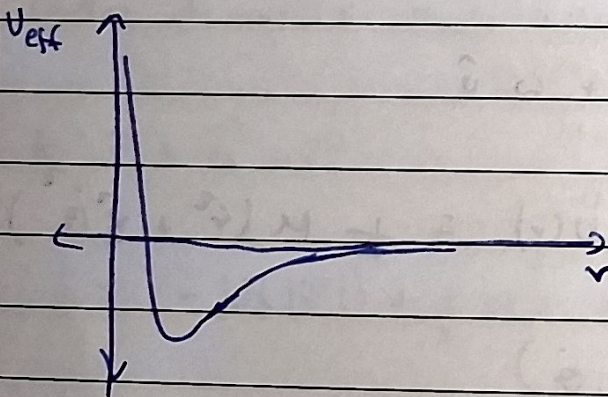
$$\frac{d}{dt} \mu \dot{r} = \mu r \dot{\theta}^2 - \frac{\partial U}{\partial r}$$

$$\mu \ddot{r} = \mu r \dot{\theta}^2 - \frac{\partial U}{\partial r}$$

$$\mu \ddot{r} = \frac{\mu l^2}{r^3} - \frac{\partial U}{\partial r} \quad (\text{from defn in the last page})$$

$$E = \underbrace{\frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\theta}^2)}_{\text{KE}} - \underbrace{\frac{G m_1 m_2}{r}}_{\text{PE}}$$

$$= \frac{1}{2} \mu \dot{r}^2 + \underbrace{\left(\frac{\mu l^2}{r^3} - \frac{G m_1 m_2}{r} \right)}_{\rightarrow U_{\text{eff}}}$$



When $E = \text{minima} = \text{circular motion}$

$E < 0 = \text{ellipsoid motion}$

$E > 0 = \text{hyperbolic motion}$

Q1 Disc rolling down an incline.

Let y be the centre of the disc:

$$\text{then } dy = R d\theta$$

$$\Rightarrow y - R\theta = 0$$

$$L = \underbrace{\frac{1}{2} m \dot{y}^2}_{\text{Linear KE}} + \underbrace{\frac{1}{2} I \dot{\theta}^2}_{\text{Rotational KE}} - \underbrace{Mg(y \sin \alpha)}_{\text{incline of the plane}} = L(y, \dot{y}, \dot{\theta})$$

$$\frac{\delta L}{\delta y} - \frac{d}{dt} \frac{\delta L}{\delta \dot{y}} + \lambda \frac{\delta f}{\delta y} = 0 = Mg \sin \alpha - m\ddot{y} + \lambda$$

$$\frac{\delta L}{\delta \theta} - \frac{d}{dt} \frac{\delta L}{\delta \dot{\theta}} + \lambda \frac{\delta f}{\delta \theta} = 0 = -I\ddot{\theta} - \lambda R$$

$$y - R\theta = 0$$

$$\text{Solving; } \ddot{y} = \left(\frac{m}{m + I/R^2} \right) g \sin \alpha$$

Continuous symmetries lead to conservation laws.

Time symmetry:- A path translated in time has no change in the Lagrangian. $\Rightarrow \frac{\delta L}{\delta t} = 0$

•

Results in Conservation of Energy.

Check Slides for derivation.

Translational symmetry:- A path translated in space has no change in the Lagrangian. $\Rightarrow \frac{\delta L}{\delta \vec{r}} = 0$

Results in the.

Conservation of Momentum

$\exists$ symmetry $\rightarrow$ something is conserved $\Rightarrow$ Noether's Theorem

For a 1-d particle (Time symmetry)

$$L = \frac{1}{2} m \dot{x}^2 - U(x)$$

$$L - \frac{d}{dq} \frac{\delta L}{\delta \dot{q}} = \left[\frac{1}{2} m \dot{x}^2 - U(x) \right] - \dot{x} (m \dot{x})$$

$$= -\frac{1}{2} m \dot{x}^2 - U(x) = -E$$

$$\text{Hamiltonian } H = \dot{q} \frac{\delta L}{\delta \dot{q}} - L = \text{Energy}$$

$$\frac{dH}{dt} = 0 \Rightarrow H \text{ is const.}$$

Space Symmetry

$$\frac{\delta L}{\delta x} = \frac{d}{dt} \frac{\delta L}{\delta \dot{x}} = 0 \Rightarrow \frac{\delta L}{\delta \dot{x}} = \text{const} = m \dot{x}$$

Hamiltonian Mechanics

From Legendre Transforms.

Deals with 1st ODE

$$H = KE + PE$$

conjugate momentum. (p)

$$H = \dot{q} \cdot \frac{\partial L}{\partial \dot{q}} - L$$

From Euler Lagrange

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q}$$

$$H = \dot{q} p - L$$

$$\frac{d}{dt} p = \frac{\partial L}{\partial q}$$

$$dH = d(\dot{q} p) - dL$$

$$= p d\dot{q} + \dot{q} dp - \left(\frac{\partial L}{\partial q} dq + \frac{\partial L}{\partial \dot{q}} d\dot{q} + \frac{\partial L}{\partial t} dt \right)$$

$$dH = \dot{q} dp - \frac{\partial L}{\partial q} dq - \frac{\partial L}{\partial t} dt$$

$$dH = \dot{q} dp - \dot{p} dq - \frac{\partial L}{\partial t} dt$$

This means that the Hamiltonian has natural variables of position, momentum & time.

$$\dot{q} = \frac{\partial H}{\partial p} \bigg|_{q,t} \quad \dot{p} = - \frac{\partial H}{\partial q} \bigg|_{p,t} \quad - \frac{\partial L}{\partial t} = \frac{\partial H}{\partial t} \bigg|_{p,q}$$

↳ Two first order differential equations.

↳ Canonical Eqs of Motion.

Q) Let $L = \frac{1}{2} m \dot{x}^2 - V(x)$

$$\frac{\partial L}{\partial \dot{x}} = p = m \dot{x} \Rightarrow \dot{x} = \frac{p}{m}$$

$$H = \frac{p^2}{m} - \frac{p^2}{2m} + U(x) = \frac{p^2}{2m} + U(x) = \text{Total mech E}$$

Canonical eqns

$$(1) \dot{x} = p/m$$

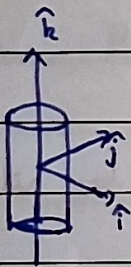
$$(2) \dot{p} = -dU/dx$$

Hamiltonian treats $\dot{q}$ & $\dot{p}$ equally

Lagrangian does not treat $\dot{q}$ & q equally

because of the $\frac{d}{dt}$ term.

Q) Particle's motion confined to a cylinder

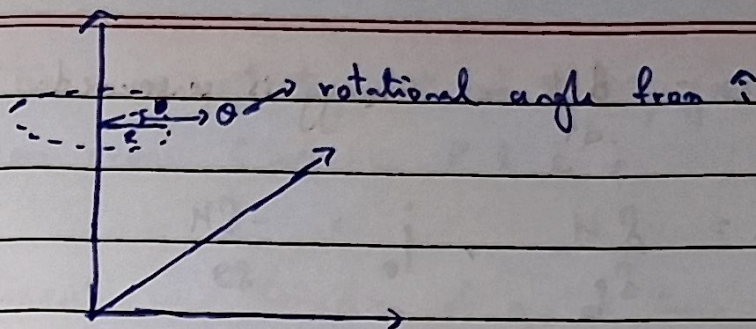


$$\vec{F}(\vec{r}) = -k\vec{r}$$

$$L = KE - PE$$

$$\vec{F} = -\nabla U = -\frac{\partial U}{\partial \vec{r}} \Rightarrow U = \frac{1}{2} k (x^2 + y^2 + z^2) \Rightarrow \frac{1}{2} k \vec{r} \cdot \vec{r}$$

$$L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - \frac{1}{2} k (x^2 + y^2 + z^2)$$



Then $x = R \cos \theta$; $y = R \sin \theta$; z

$$L(\theta, z, \dot{\theta}, \dot{z}, t) = \frac{1}{2} m (R^2 \dot{\theta}^2 + \dot{z}^2) - \frac{1}{2} k (R^2 + z^2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = \frac{\partial L}{\partial \theta} \quad \& \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z}$$

$$H = \dot{q} p - L = \dot{\theta} p_{\theta} + \dot{z} p_z - L$$

$$p_{\theta} = \frac{\partial L}{\partial \dot{\theta}} = m R^2 \dot{\theta} \quad (\text{defined}) \Rightarrow \dot{\theta} = p_{\theta} / m R^2$$

$$p_z = \frac{\partial L}{\partial \dot{z}} = m \dot{z} \quad (\text{defined}) \Rightarrow \dot{z} = p_z / m$$

$$H(\theta, z, p_{\theta}, p_z, t) = \frac{p_{\theta}^2}{m R^2} + \frac{p_z^2}{m} - \frac{1}{2} m \left(R^2 \frac{p_{\theta}^2}{m^2 R^2} + \frac{p_z^2}{m^2} \right) + \frac{1}{2} k (R^2 + z^2)$$

$$= \frac{1}{2} \frac{p_{\theta}^2}{m R^2} + \frac{1}{2} \frac{p_z^2}{m} + \frac{1}{2} k (R^2 + z^2)$$

$$= KE + PE$$

$$\frac{\partial H}{\partial t} = 0 = \frac{\partial H}{\partial t} \Rightarrow \text{Energy is conserved.}$$

$$\dot{q}_0 = \dot{q} = \frac{\partial H}{\partial p_0}, \quad \dot{p}_0 = -\frac{\partial H}{\partial q}$$

$$\dot{z} = \frac{\partial H}{\partial p_z}, \quad \dot{p}_z = -\frac{\partial H}{\partial z}$$

Here; H is a fn of q, p_0, p_z, z , hence

$$\frac{\partial H}{\partial q} = 0 \Rightarrow \dot{p}_0 = 0 \Rightarrow p_0 \text{ is const} = \dot{q} \text{ is const}$$

$$= 0 = \alpha t$$

$$\dot{p}_z = k_z.$$

Phase Space:-

Defined by the vector $(q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n)$

$$\text{Let } \dot{q}_k = \frac{\partial H}{\partial p_k}; \quad \dot{p}_k = -\frac{\partial H}{\partial q_k}; \quad k = 1, \dots, n$$

Take the simplified case with q & p

$$\dot{q} = \frac{\partial H}{\partial p} \quad \& \quad \dot{p} = -\frac{\partial H}{\partial q}$$

This ~~is~~ restricts that no two trajectories of a particle can cross (just like $\vec{E}$ & $\vec{B}$)

Ensemble:- Collection of q_s & their respective p_s in m^2 space.

Under Hamiltonian Dynamics, this blob cannot change volume - Liouville's Theorem.

Say your blob is a rectangle of $dq \times dp$
After any Δt ; $dq \times dp$ is still the same.

Let $\rho(p, q, t)$ is a probability density fn.

$$\int dq dp \rho(p, q, t) = 1$$

Boltzmann derived $\rho = e^{-\beta H(p, q)}$

Check proof for Liouville's Theorem

Continuity Eqn:

Special Theory of Relativity.

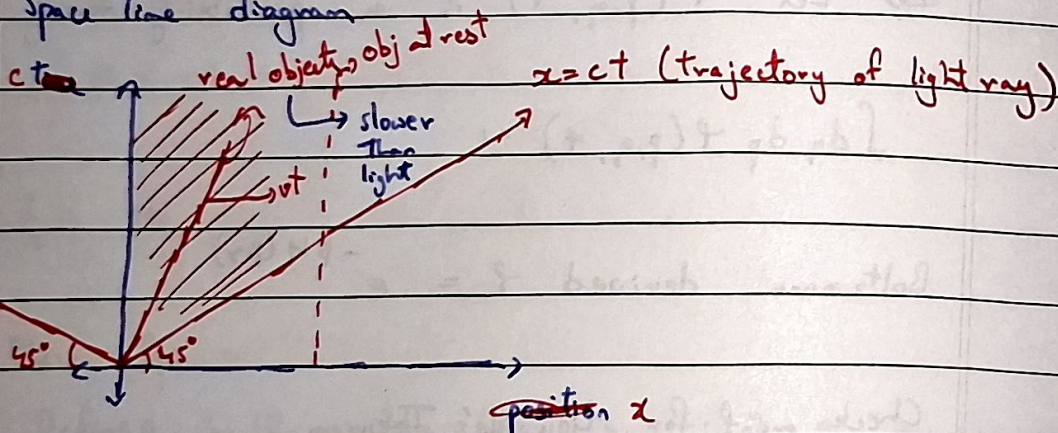
Frames:- Inertial & non-inertial frames

The forces acting on a body must be the same across any inertial frame of reference

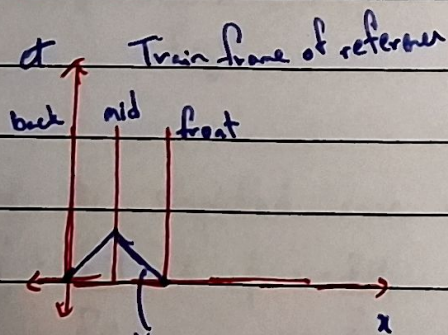
STR deals with inertial frames only.

What is simultaneous in one inertial frame need not be simultaneous in another inertial frame.

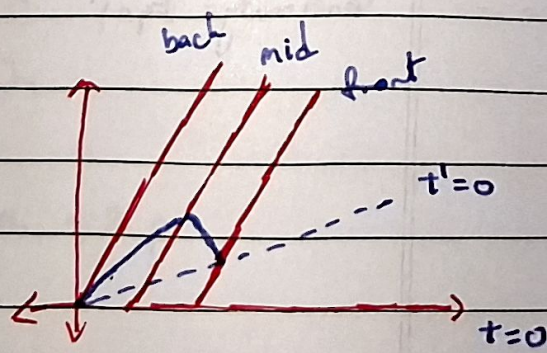
Space Time diagram



For a train



platform frame of reference



t' is a different frame of reference to t .

$$x' = \gamma(x - vt)$$

$$x = \gamma(x' + vt')$$

$$y' = y$$

$$\Rightarrow$$

$$y = y'$$

$$z' = z$$

$$z = z'$$

$$t' = \gamma\left(t - \frac{xv}{c^2}\right)$$

$$t = \gamma\left(t' + \frac{x'v}{c^2}\right)$$

Lorentz Transform

Inverse Lorentz Transform



$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

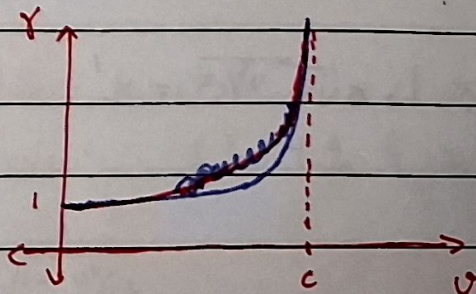
Converts from K-frame to a K'-frame that is moving with constant v ^{along} one dimension wrt K-frame.

Einstein insisted that γ must be the same on both sides and adding v to one side $\Leftrightarrow$ subtracting v on the other side.

When $v \rightarrow 0$; $\gamma \rightarrow 1$.

Taylor expanding γ when $v \rightarrow 0$ (or $\frac{v}{c} \ll 1$)

$$\gamma(v) = 1 + \frac{1}{2} \frac{v^2}{c^2}$$



Space-Time Invariant is same in both frames

Show This?

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$$

Time Dilation:-

Take the events

$$e_0 = (0, 0); e_1 = (0, c\tau_0); e_2 = (0, 2c\tau_0); \dots$$

in the K -frame

Using Lorentz Transform

$$e'_0 = (0, 0); e'_1 = (-v\gamma c\tau_0, c\tau_0\gamma); \dots$$

Generally

$$e_n = (-\gamma v n c\tau_0, \gamma n c\tau_0)$$

Time difference b/w 2 ticks in the K' -frame

$$\gamma(n(n-1))c\tau_0 = \gamma c\tau_0$$

Length contraction

Let the rod be on the platform with ends

$$e_1 = (0, t); e_2 = (L_0, t)$$

We need to find e'_1 & e'_2 with $t'_1 = t'_2$

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$

$$\text{Solving; } L_0 \sqrt{1 - v^2/c^2} = L'$$

Galilean Relativity:-

$$x' = x - vt \quad ; \quad t' = t$$

$$\frac{dx'}{dt} = \frac{dx}{dt} = \frac{dv}{dt} = 0$$

$$u' = u - v$$

STR:- constant speed of light $\Rightarrow t' = t$

Leads to the concept of relativity of simultaneity

Setup:- For Lorentz Transform

K-frame $\rightarrow$ Earth fixed frame

K'-frame $\rightarrow$ aligned x-axes, origin moves along x-axis with relative velocity v . Coincident frames at $t=0$

Location of origin $O' \Rightarrow \overset{K}{\uparrow} x - vt = 0 \quad \& \quad \overset{K'}{\uparrow} x' = 0$

$\Rightarrow x' = \gamma(x - vt)$ assuming linear transformations

First postulate requires $x = \gamma'(x' + vt')$ & $\gamma = \gamma'$

Eliminate x' and rearrange to get

$$t' = \gamma t + \frac{x}{\gamma v} (1 - \gamma^2)$$

From second postulate; $x = ct$ & $x' = ct' \Leftarrow$ wavefront of light has the same velocity. Substitute & simplify to

acquire
$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$ct' = \gamma((c-v)t)$$

Measurements happen by observers who are local (at the experiment) such that $t_{\text{light}} = 0$.

This is not possible, so we must correct for it.

Doppler Effect

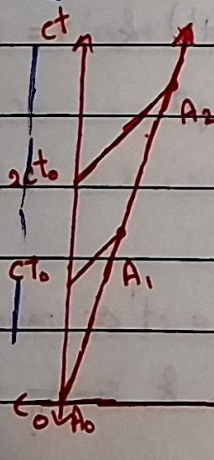
Apparent change in the frequency of the sound (and light) due to the motion of the source.

STR says that c cannot change, but the Doppler effect is still present.

$$\text{let } \beta = v/c$$

$$\nu' = \frac{1}{\Delta t'} = \frac{\nu}{\sqrt{\frac{1-\beta}{1+\beta}}}$$

Assume stationary light source which emits light every t_0 seconds.



$$A_0 = A_0' = (0, 0)$$

$$A_1: ct = x + ct_0 \text{ \& } x = vt$$

$$\Rightarrow ct_1 = t_0/(c-v)$$

$$\text{Similarly } A_2: ct_2 = 2t_0/(c-v)$$

$$A_1 = \left(\frac{vct_0}{c-v}, \frac{ct_0}{c-v} \right) \Rightarrow A'_1 = \left(0, \gamma \left[\frac{ct_0}{c-v} - \frac{v}{c^2} \frac{vct_0}{c-v} \right] \right)$$

$$\Delta t' = \frac{\Delta t}{\gamma} = \Delta t \sqrt{\frac{1-\beta}{1+\beta}}$$

Deriving Relative Velocity Formulae.

$x(t) = ut$ in K^0 frame.

$$x' = \gamma(x - vt)$$

$$\frac{dx'}{dt'} = \frac{dx'/dt}{dt'/dt} = \frac{\gamma \left(\frac{dx}{dt} - v \right)}{\gamma \left(1 - \frac{v}{c^2} \frac{dx}{dt} \right)} = \frac{u_x - v}{1 - u_x v/c^2}$$

$u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}} \Rightarrow$ small velocities of the object & small relative velocities $\leftarrow$ we get back the (?) formula.

$$\frac{dy'}{dt'} = \frac{dy'/dt}{dt'/dt} = \frac{dy/dt}{\gamma \left(1 - \frac{v u_x}{c^2} \right)} = \frac{u_y}{\gamma \left(1 - \frac{u_x v}{c^2} \right)} = u_y'$$

Similarly

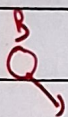
$$\frac{dz'}{dt'} = \frac{u_z}{\gamma \left(1 - \frac{u_x v}{c^2} \right)} = u_z'$$

$$u_x' \Big|_{u_x=c} = \frac{c-u}{1-\frac{u}{c}} = c$$

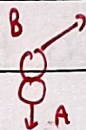
But what about Relativistic Momentum?

Newtonian $\vec{p} = m\vec{v}$. But this is insufficient

Assume 2 carbon coins



$\Rightarrow$



$\hat{O}A$

$$u_x^A = 0$$

$$u_x^A = 0$$

$$u_y^A = u_y$$

$$u_y^A = -u_0$$

$$u_x^B = u_1$$

$$u_x^B = u$$

$$u_y^B = u_2$$

$$u_y^B = -u_2$$

Assume a K' frame with $v = u_1$
then ball B moves only vertically

Check slides - little lost